Corso: Data science lab: process and methods
Quiz: Exam (2025-01-24)

PORTALE ESAMI

	Nunzio Licalzi **s344860**
Iniziato	24 gennaio 2025, 08:41
Stato	Completato
Terminato	24 gennaio 2025, 10:11
Tempo impiegato	1 ora 30 min.
Valutazione	**18,75** su un massimo di 20,00 (**94**%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12 13

Domanda 1

Risposta corretta

Punteggio ottenuto 2,00 su 2,00

2 points (-15% penalty for a wrong answer)

- precision(C) is computed as the fraction of correctly predicted points belonging to class C, with respect to all points predicted as belonging to class C.
- recall(C) is computed as the fraction of correctly predicted points belonging to class C, with respect to all points actually belonging to class C.
- F1(C) is computed as the harmonic mean between precision(C) and Recall(C).

The following are the definitions of weighted and macro precision. We assume a classification problem with M classes (1, 2, ..., M), and N samples to be classified. We assume the cardinality for class i as being n_i (in other words, $\sum_{i=1}^{M} c_i = N$).

$$\text{weighted precision} = \sum_{i=1}^{M} \frac{c_i}{N} \text{precision}(i)$$
$$\text{macro precision} = \frac{1}{N} \sum_{i=1}^{M} \text{precision}(i)$$

In other words, the weighted precision is the average precision across all classes, weighted by the number of samples belonging to each class, whereas the macro precision is the unweighted average of all precisions.

Weighted/macro recall/F1 can be computed similarly, replacing precision(i) with recall(i) and F1(i).

What is the weighted recall equal to?

- a. Precision(1) = Precision(2) = ... = Precision(N)
- b. None of the other answers is correct
- c. Macro F1
- d. F1(1) = F1(2) = ... = F1(N)
- e. The provided information is not sufficient to answer the question
- f. Macro recall
- g. Accuracy ✔
- h. Recall(1) = Recall(2) = ... = Recall(N)
- i. Weighted precision
- j. Weighted F1 score
- k. Macro precision

Your answer is correct.

La risposta corretta è:
Accuracy

Domanda 2
Risposta corretta
Punteggio ottenuto 1,50 su 1,50

1.5 points (-15% penalty for each wrong answer)

Which of the following algorithms is incremental? Choose all correct answers (multiple answers may be correct).

- [ ] a. K-means
- [ ] b. SVM
- [x] c. Naive Bayes ✔
- [ ] d. Hierarchical (agglomerative) clustering
- [ ] e. DBSCAN
- [ ] f. Association rules
- [ ] g. Decision Tree
- [ ] h. Random Forest
- [x] i. K-Nearest Neighbors ✔

Your answer is correct.

Le risposte corrette sono:
Naive Bayes,

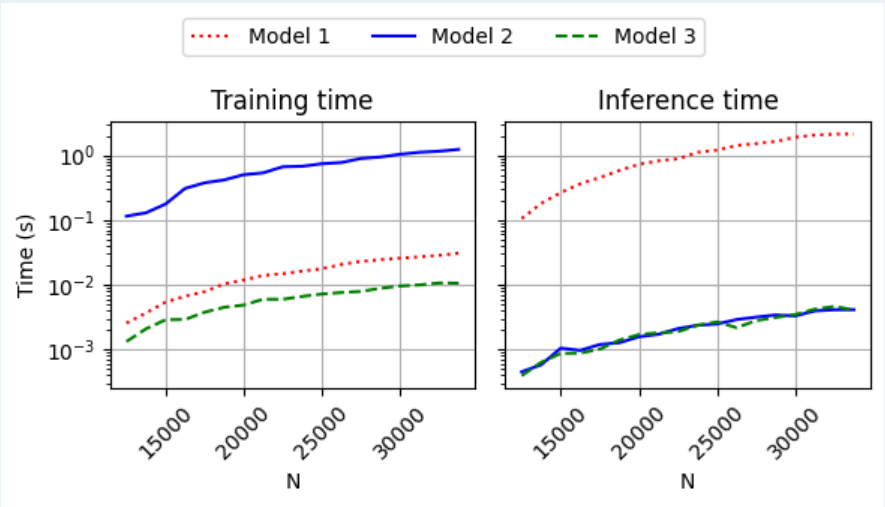
K-Nearest Neighbors

Domanda 3
Risposta corretta
Punteggio ottenuto 2,00 su 2,00

2 points (-15% penalty for a wrong answer)

Three classification algorithms have been adopted to address a classification problem: Decision Tree, K-Nearest Neighbors, and Naive Bayes.

The following two plots report the training and inference times for the three models, as the number of samples increases (note that the same line style is used to visualize the same model in both graphs).



Which of the following is the correct model assignment?

- [ ] a. Model 1 = K-Nearest Neighbors
  Model 2 = Naive Bayes
  Model 3 = Decision Tree

b. The provided information is insufficient to answer the question

c. Model 1 = Naive Bayes
Model 2 = K-Nearest Neighbors
Model 3 = Decision Tree

d. Model 1 = K-Nearest Neighbors ✔
Model 2 = Decision Tree
Model 3 = Naive Bayes

e. Model 1 = Decision Tree
Model 2 = Naive Bayes
Model 3 = K-Nearest Neighbors

f. Model 1 = Decision Tree
Model 2 = K-Nearest Neighbors
Model 3 = Naive Bayes

g. Model 1 = Naive Bayes
Model 2 = Decision Tree
Model 3 = K-Nearest Neighbors

Your answer is correct.

La risposta corretta è:
Model 1 = K-Nearest Neighbors
Model 2 = Decision Tree
Model 3 = Naive Bayes

Domanda 4
Risposta corretta
Punteggio ottenuto 2,00 su 2,00

2 points (no penalty for a wrong answer)

For a dataset with target variables $y = (y_1, y_2, \ldots, y_n)$ and predictions $\hat{y} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n)$, the R2 is defined as:

$$R2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Where $\bar{y}$ is defined as the mean value of y.

A test set comprised of 3 points is given. For each point, the two input features (x1, x2) and the continuous target (y) are known.

A total of 6 different models are used to produce predictions for the 3 points (pred_1, pred_2, ..., pred_6).

The following table summarises the abovementioned information.

x1	x2	y	pred_1	pred_2	pred_3	pred_4	pred_5	pred_6
1.10	-0.47	51.46	74.61	69.40	68.28	58.15	50.55	69.20
-0.12	0.78	34.26	45.88	55.96	48.79	28.75	36.01	49.72
0.97	1.17	130.52	156.83	167.02	160.66	128.56	137.83	155.48

What is the R2 score of the model having the best performance on the test set?

Write the answer in the box below.

Write the result as a decimal number, using at least 4 significant figures.

Risposta: 0.98912 ✔

La risposta corretta è : 0,98912365827963

Domanda 5

Risposta corretta

Punteggio ottenuto 1,50 su 1,50

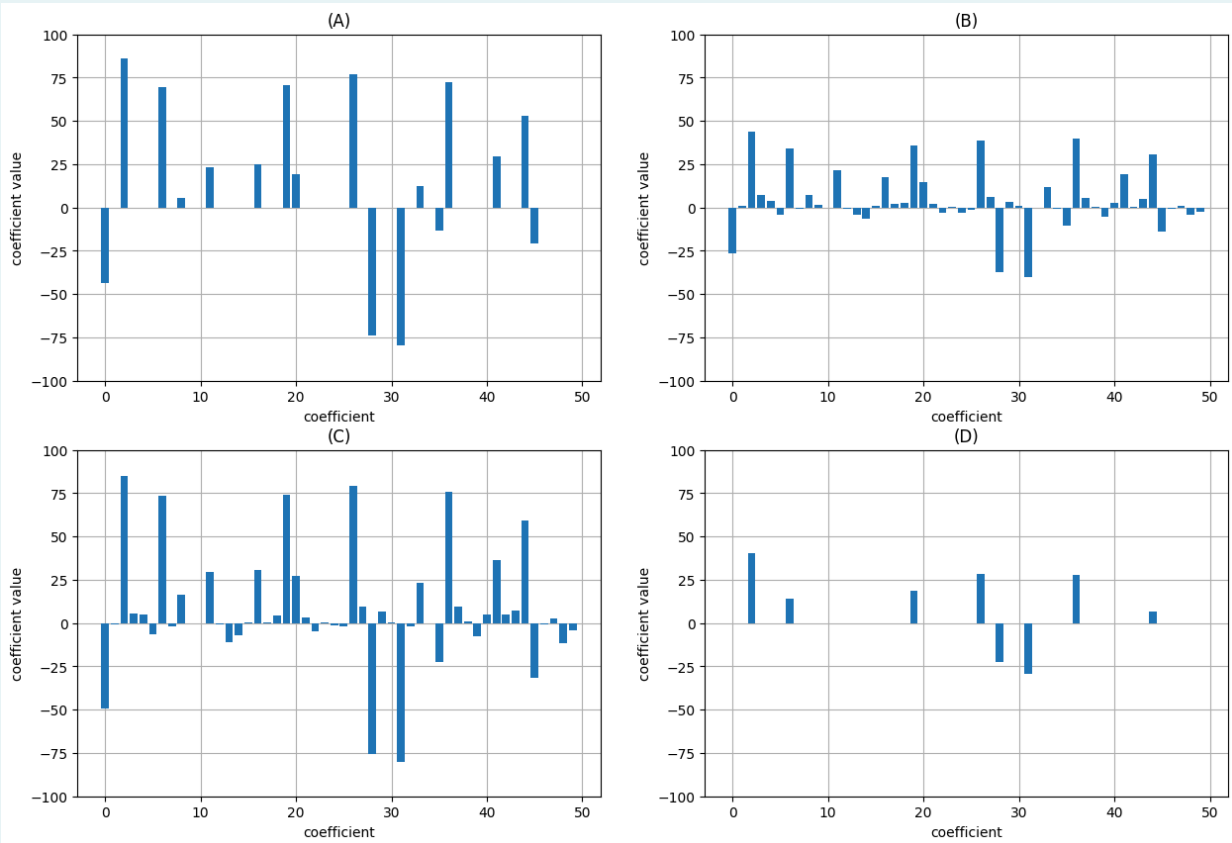
1.5 points (-15% penalty for a wrong answer)

The weights w of a Ridge model are chosen so as to minimize $MSE + \lambda||w||_2^2$
The weights w of a Lasso model are chosen so as to minimize $MSE + \lambda||w||_1$

Two Ridge models and two Lasso models have been trained on the same dataset comprised of 1,000 50-dimensional points. In particular, the following are the characteristics of the four models:

- Ridge, $\lambda = 10$
- Ridge, $\lambda = 100$
- Lasso, $\lambda = 1$
- Lasso, $\lambda = 5$

After training, you plot the weights (coefficients) learned by the four models as bar charts. The following are the plots you obtain.



You labelled the four plots with letters (A, B, C, D respectively). Which model corresponds to which plot?

Select all correct mappings. Each plot is associated to only one correct answer.

- [ ] a. (A) Lasso, $\lambda = 5$
- [ ] b. (A) Ridge, $\lambda = 100$
- [x] c. (B) Ridge, $\lambda = 100$ ✔
- [x] d. (D) Lasso, $\lambda = 5$ ✔
- [ ] e. (D) Ridge, $\lambda = 10$
- [ ] f. (B) Ridge, $\lambda = 10$
- [ ] g. (B) Lasso, $\lambda = 5$

- [ ] h. (C) Lasso, $\lambda = 1$
- [ ] i. (D) Lasso, $\lambda = 1$
- [ ] j. (C) Lasso, $\lambda = 5$
- [ ] k. (B) Lasso, $\lambda = 1$
- [x] l. (A) Lasso, $\lambda = 1$ ✔
- [x] m. (C) Ridge, $\lambda = 10$ ✔
- [ ] n. (C) Ridge, $\lambda = 100$
- [ ] o. (D) Ridge, $\lambda = 100$
- [ ] p. (A) Ridge, $\lambda = 10$

Your answer is correct.

Le risposte corrette sono:
(A) Lasso, $\lambda = 1$

,
(B) Ridge, $\lambda = 100$

,
(C) Ridge, $\lambda = 10$

,
(D) Lasso, $\lambda = 5$

Domanda 6
Completo
Punteggio ottenuto 1,50 su 1,50

1.5 points (no penalty for a wrong answer)

You are given the following distance matrix among 5 points, a, b, c, d, e.

	a	b	c	d	e
a	0	18	10	5	14
b	18	0	2	17	4
c	10	2	0	24	6
d	5	17	24	0	9
e	14	4	6	9	0

Apply hierarchical clustering using complete (MAX) linkage.

Write in the box below the clusters obtained at each step of the clustering.

Use a separate line to represent the state after each step. Separate the clusters obtained using a vertical bar (|).

Start from the case where each point belong to its own cluster. End with the case where all points belong to a single cluster.

For example,

a | b | c | d | e
a b | c | d | e
a b c | d | e
a b c d | e
a b c d e

In this case, the steps of the clustering first merge a with b, then (a, b) with c, then (a, b, c) with d, and finally (a, b, c, d) with e.

```
a | b | c | d | e
a | b c | d | e
a d | b c | e
a d | b c e
a d b c e
```

a|b|c|d|e
a|d|e|bc
e|bc|ad
ad|ebc
abcde

Commento:
ok

Domanda 7
Completo
Punteggio ottenuto 2,50 su 2,50

2.5 points (no penalty for a wrong answer)

You are given the following list of transactions.

```
ae
cde
abcd
abc
acde
bce
abce
ade
c
cde
```

Apply the FP-growth algorithm using minsup = 2 (an itemset is frequent if it appears in at least two transactions).

Write the following:

1. B-CPB (B - Conditional Pattern Base)
2. B-CHT (B - Conditional Header Table)
3. BE-CPB (BE - Conditional Pattern Base)

Write all and only the correct itemsets, along with their support counts.

Use the following syntax:

B-CPB = { element1: support1, element2: support2, ... }
B-CHT = { element1: support1, element2: support2, ... }
BE-CPB = { element1: support1, element2: support2, ... }

For example:

B-CPB = { a: 1, abc: 2 }
B-CHT = { a: 3, b: 2, c: 1 }
BE-CPB = { a: 1, b: 2, c: 3 }

```
B-CPB = {CAD:1, CA:1, CEA:1, CE:1}
B-CHT = {C:4, A:3, E:2, D:1}
BE-CPB = {CA:1, C:1}
```

Commento:
B-CHT
Extra itemsets: {d} not frequent

Domanda 8

Risposta corretta

Punteggio ottenuto 1,50 su 1,50

1.5 points (-15% penalty for a wrong answer)

A frequent itemset is closed if none of its supersets has its same support.
A frequent itemset is maximal if none of its supersets is frequent.

The following are the frequent itemsets that have been extracted from a transactional database (reported as *itemset: support*).

D: 4
B: 4
A: 4
BD: 3
AD: 3
AB: 2
ABD: 2

Which of the following statements is correct?

- [ ] a. BD is not closed
- [ ] b. None of the other statements is correct
- [ ] c. D is maximal
- [ ] d. AD is not closed
- [ ] e. D is not closed
- [ ] f. A is not closed
- [ ] g. The provided information is insufficient to determine the correctness of the other statements
- [ ] h. ABD is not closed
- [x] i. AD is closed, but not maximal ✔

Your answer is correct.

La risposta corretta è: AD is closed, but not maximal

Domanda 9

Completo

Punteggio ottenuto 1,75 su 2,00

2 points (no penalty for a wrong answer)

The following are the contents of pandas DataFrame *df*.

	age	score	group
0	2	1	4
1	3	3	2
2	4	4	1
3	1	2	2
4	4	2	4
5	1	4	3
6	4	1	3
7	4	1	2
8	1	2	2
9	3	2	3

What is the output of this code snippet?

```
grouped = df.groupby("group")
filtered_df = grouped.filter(lambda x: x["score"].mean() > 3)
out = filtered_df.groupby("group")[["age", "score"]].mean()
print(out)
```

Write the answer in the box below, specifying:

- Type (DataFrame or Series)
- Columns (for DataFrames only)
- Index
- Data (the contents of the Series/DataFrame)

If the execution of the code results in an error, write “an error occurs” instead.

For "Data", you can separate multiple columns (in the case of DataFrames) using a space. You can separate rows using newlines. **Use decimal numbers, with at least 4 significant figures. Fraction notations (Numerator/Denominator) will not be considered valid answers.**

In the case of a multi-level index/column, you must represent them in the same way as with the data (separating with spaces and newlines). You need to repeat the outermost index/column whenever needed.

The following are some examples of the representation you should use.

Example 1

DataFrame

	a	b	c
0	3	3	2
1	2	4	4

Expected representation

```
Type
DataFrame

Columns
a b c

Index
0 1

Data
3 3 2
2 4 4
```

Example 2

Multi-index/column DataFrame

		W		X
		a	b	c
y	0	2	3	3
	1	4	3	4
z	0	3	1	4
	1	3	3	4

Expected representation

```
Type
DataFrame

Columns
W W X
a b c

Index
y y z z
0 1 0 1

Data
2 3 3
4 3 4
3 1 4
3 3 4
```

Example 3

Series

a	11
b	7
c	5
d	−1

Expected representation

```
Type
Series

Index
a b c d

Data
11 7 5 -1
```

```
Type:
Series

Index:
age score

Data:
4.000000 4.000000
```

Commento:

Type
DataFrame

Index
1 (group)

Columns
age score

Data
4 4

Domanda
10

Risposta corretta

Punteggio ottenuto 1,50 su 1,50

1.5 points (no penalty for a wrong answer)

What is the output of the following snippet of code?

```
1  a = np.random.random((100, 32, 32))
2  b = np.random.random((100, 32, 32, 3))
3  c = b.sum(axis=3).max(axis=2)
4  d = np.vstack([a.sum(axis=1), c])
5  print(d.shape)
```

Write the answer in the box below. If an error occurs, write "Error at line #" -- where # is the line number.

Enclose the answer in paranthesis ().

Separate each dimension with a comma (no spaces).

Failure to do so will result in an incorrect evaluation of the answer.

Examples of answers

For a 2-dimensional matrix with 2 rows, 3 columns, write:

(2,3)

For a 1-dimensional row vector with 5 elements, write:

(5) or (5,)

For a 1-dimensional column vector with 7 elements, write:

(7,1)

For an array with shape 3x2x9x11, write:

(3,2,9,11)

If an error occurs at line 12, write:

Error at line 12

Risposta: (200,32)

La risposta corretta è : (200,32)

Domanda 11

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

1 point (no penalty for a wrong answer)

For two N-dimensional vectors x and y, the cosine distance is defined as:

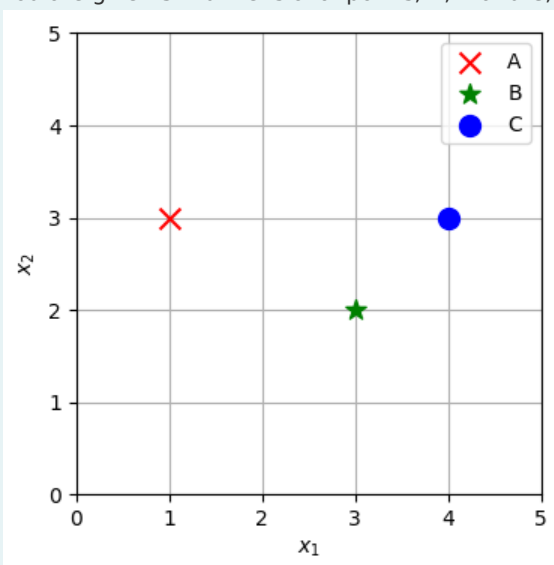
$$d_{COS}(x,y) = 1 - \frac{x \cdot y}{||x||||y||}$$

where $x \cdot y$ is the dot product of x and y, and $||x||$ is the norm of x.

The Manhattan distance is defined as:

$$L1(x,y) = \sum_{i=1}^{N} |x_i - y_i|$$

You are given 3 2-dimensional points, A, B and C, as shown below.



Compute the Manhattan distance between A and C, and the cosine distance between A and B.

Write the answers in the corresponding boxes. **Write the answers as decimal numbers, using at least 4 significant figures.**

$L1(A,C) =$ 3.000000 ✔

$d_{COS}(A,B) =$ 0.21065 ✔

Domanda 12

Risposta non data

Punteggio max.: 1,00

1 point (-15% penalty for a wrong answer)

Which of the following sequences of scikit-learn estimators does not constitute a valid pipeline?

- a. SimpleImputer, PolynomialFeatures, LinearRegression
- b. MinMaxScaler, PCA, KMeans
- c. None of the other options is correct
- d. All of the other options are valid pipelines in scikit-learn
- e. SimpleImputer, StandardScaler, RandomForestClassifier

- f. DecisionTreeClassifier, StandardScaler
- g. StandardScaler, PCA, KNeighborsClassifier
- h. CountVectorizer, DecisionTreeClassifier

Your answer is incorrect.

La risposta corretta è:
DecisionTreeClassifier, StandardScaler

Domanda
13

Risposta non data

Non valutata

This is NOT a question

You can use this edit box for any note you may want to take or any draft of your solutions. If you'd like to leave any notes for the professors, you can write them here. No points will be assigned to this question.